A Tutor That Reads the Room

Infers mastery, frustration, and engagement — and advises the next move.

ADVISORY

— THE PROBLEM

The material isn't the problem — the timing is

- Push a frustrated learner harder and they quit.
- Coast a strong learner and they disengage.
- A fixed difficulty schedule is wrong for almost everyone at once.

`Good tutoring is reading the learner's state and meeting it.`

— THE SOLUTION

Read the signals a human tutor reads

- **It senses**
  answer correctness, response time, and hint requests per topic.
- **It infers**
  mastery, frustration, and engagement — the learner's actual state.
- **It advises**
  a hint and easier work, or a harder exercise — to the platform.

— HOW IT WORKS

Learner events in, a recommendation out

0 **Learner event**
correctness · time · hint · topic

↓

1 **Feature**
mastery · frustration · engagement

↓

2 **Policy / advisory**
difficulty · hint

↓

3 **Result**
tutor advisory

A feature layer estimates state; a policy layer chooses the next move.

— THE PROOF

Same model, opposite advice

STRUGGLING LEARNER

- Repeated wrong answers, long response times.
- → offer a hint, lower the difficulty.

STRONG LEARNER

- Fast, correct answers.
- → raise the difficulty, harder exercise.

— THE CEILING

Advisory, educator-led

ADVISORY Recommends a difficulty change or a hint; the platform and educator decide.

Learner-aware Tracks the individual's state, not a one-size schedule.

Auditable Every advisory carries the state that produced it.

— GOING LIVE

Plug into the LMS you already use

- **Swap the input edge**
  the mock events become a live LTI / xAPI source.
- **Keep the contract**
  same typed signals, same advisory ceiling.
- **Keep the educator**
  a human stays in charge of the pedagogy.

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Adaptive learning works when it meets the student where they are — and leaves the teaching to the teacher.

The case for Demo 08

JNEOPALLIUM · DEMO 08

rakovpublic/jneopallium

Meet every learner where they are.

Mastery-and-affect-aware tutor advisories — the platform stays in control.

github.com/rakovpublic/jneopallium

ADVISORY · deterministic